



UrbanFix

Local Skilled Worker Booking System

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Slide Overview

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Project Overview

UrbanFix is a comprehensive web-based platform that bridges the gap between skilled workers and customers in need of their services.

What is it?

- Digital marketplace for skilled labor services
- Centralized platform for worker discovery
- Real-time booking and management system
- Quality assurance through profiles and tracking

Why UrbanFix?

- Solves urban labor market inefficiencies
 - Provides transparency and reliability
 - Enables professional worker management
 - Creates seamless customer experience
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2 Problem Statement

Traditional Challenges:

- ✗ Finding reliable skilled workers is difficult
- ✗ No centralized platform for booking
- ✗ Lack of transparency in pricing
- ✗ No quality assurance mechanism
- ✗ Workers have no digital presence
- ✗ No scheduling efficiency

Impact:

Customers waste time and money, workers lack opportunities, and the market remains unorganized and inefficient.

3 Solution & Objectives

Our Solution:

Create a digital platform that connects workers and customers efficiently

Project Objectives:

- ✓ Develop user-friendly booking platform
 - ✓ Implement secure authentication
 - ✓ Create efficient worker search system
 - ✓ Establish reliable booking & scheduling
 - ✓ Provide admin tools for management
 - ✓ Ensure data security & privacy
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4 User Types

Three Main User Categories:

Customers

- Browse and search for workers
- Book services based on budget

- Manage bookings and profiles
- Contact workers and support

Workers

- Register and showcase skills
- Manage availability
- Receive booking requests
- Update service details

Administrators

- Oversee platform operations
- Manage users and workers
- Handle disputes and quality
- View analytics and reports

5 Service Categories

Six Core Service Categories:

1. ⚡ **Electrician** - Electrical installations and repairs
2. 🪚 **Carpenter** - Woodwork and furniture services
3. ❄️ **AC Repairing** - Air conditioning maintenance
4. 🎨 **Painting** - Interior and exterior painting
5. 🛠️ **Mechanics** - Automotive and mechanical repairs
6. 🚰 **Plumber** - Plumbing installations and maintenance

Each category has dedicated workers with specific expertise.

6 Key Features - Part 1

Authentication & User Management:

- ✅ Secure registration and login
- ✅ Email verification system
- ✅ Separate login for customers & workers
- ✅ Password encryption with bcrypt
- ✅ JWT token-based authentication

Worker Search & Discovery:

- ✅ Category-wise filtering
- ✅ Budget-based worker selection
- ✅ Browse all workers
- ✅ View detailed profiles

7 Key Features - Part 2

Booking System:

- ✓ Real-time booking confirmation
- ✓ Track active bookings
- ✓ Historical booking records
- ✓ Request management for workers
- ✓ Instant service confirmation

User Profiles:

- ✓ Customer profile management
 - ✓ Booking history tracking
 - ✓ Worker skill management
 - ✓ Availability updates
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8 Dashboard Features

Customer Dashboard:

- View all bookings (active & completed)
- Worker profile details
- Booking history
- Request status tracking

Worker Dashboard:

- Incoming booking requests
- Manage availability
- View customer details
- Track earnings

Admin Dashboard:

- User management
 - Platform analytics
 - Quality monitoring
 - System oversight
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9 Technology Stack - Frontend

Frontend Technologies:

- **React.js** - Component-based UI framework
- **Vite** - Fast build tool
- **Tailwind CSS** - Utility-first styling
- **Bootstrap** - Responsive components
- **Framer Motion** - Smooth animations
- **React Router** - Navigation
- **HTML5 & CSS3** - Semantic markup
- **JavaScript (ES6+)** - Client-side logic

Why These?

Fast loading, responsive design, smooth animations, better UX.

10 Technology Stack - Backend & Database

Backend Technologies:

- **Node.js** - JavaScript runtime
- **Express.js** - Web framework
- **REST API** - Standard API architecture
- **Nodemailer** - Email notifications

Database:

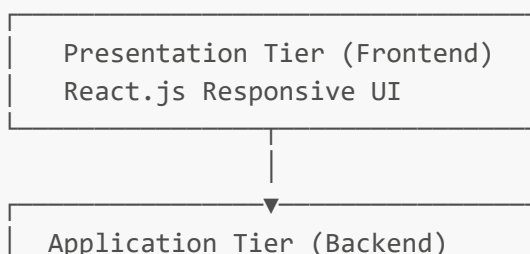
- **MongoDB** - NoSQL database
- **Mongoose** - Object modeling

Security & Validation:

- **JWT** - Token authentication
- **Bcrypt/Bcryptjs** - Password hashing
- **Joi** - Input validation
- **CORS** - Security headers

1 1 System Architecture

Three-Tier Architecture:





1 2 API Routes & Integration

Main API Routes:

Route	Purpose
/auth	User registration & login
/workers	Worker profiles & management
/admin	Administrative functions
/contact	Contact form & emails

Features:

- CORS enabled for security
- JSON request/response
- Error handling
- Email notifications via Nodemailer

1 3 Security & Authentication

Security Measures:

✔ Password Security

- Bcrypt encryption (rounds: 10)
- No plaintext storage

✔ Authentication

- JWT tokens with expiry
- Role-based access control

✔ Input Validation

- Joi schema validation
- Request sanitization

✓ API Security

- CORS protection
 - Rate limiting ready
 - Environment variables for secrets
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1 4 Current Status & Roadmap

✓ Completed Features:

- User authentication system
- Worker search & filtering
- Real-time booking system
- Dashboards (Customer, Worker, Admin)
- Contact page with emails
- Category browsing

🚀 In Development:

- Payment gateway integration
 - Review & rating system
 - Advanced notifications
 - Analytics & reporting
 - Hourly pricing implementation
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1 5 Conclusion & Impact

Project Impact:

- 🎯 Connects skilled workers with customers efficiently
- 🎯 Provides transparent pricing system
- 🎯 Ensures quality through profiles & tracking
- 🎯 Creates professional platform for workers
- 🎯 Revolutionizes urban labor market

Future Vision:

Scale to other cities, add payment systems, implement AI-based worker matching, and create mobile applications.

UrbanFix - Making skilled labor accessible to everyone!
